



UNIVERSITY OF GHANA

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BSc. COMPUTER SCIENCE

SUPPLEMENTARY EXAMINATION: 2024/25

DEPARTMENT OF COMPUTER SCIENCE

DCIT308: DATA STRUCTURES AND ALGORITHMS II (3 CREDITS)

INSTRUCTION:

Please ensure that all your answers are brief and properly demonstrate your understanding of the concepts. Marks will be awarded for clarity of expression. Answer all questions in the answer booklet provided.

Answer Three (3) Questions out of the Four (4) Questions

TIME ALLOWED:

TWO (2) HOURS

ANSWER THREE (3) QUESTIONS IN ALL

1.

(a) Explain the difference between an array and a linked list in terms of memory allocation and access time. (5 points)

(b) Given an unsorted array of integers, describe an efficient algorithm to search for a specific element. Compare its efficiency with searching in a linked list. (7 points)

(c) Write a function (in any programming language) to reverse a singly linked list. (8 points)

TOTAL = [20 MARKS]

2.

(a) What is the main difference between a stack and a queue? Provide an example of a real-world application for each. (5 points)

(b) Implement a queue using two stacks and explain the time complexity of the enqueue and dequeue operations. (7 points)

(c) Explain how a circular queue works and how it improves efficiency over a regular queue. Provide an example where it is useful. (8 points)

TOTAL = [20 MARKS]

3.

(a) Define a binary search tree (BST) and explain its key properties. (5 points)

(b) Given a binary tree, describe an algorithm to find its height. (7 points)

(c) Discuss the difference between Depth-First Search (DFS) and Breadth-First Search (BFS) in graph traversal, including their time complexities. (8 points)

TOTAL = [20 MARKS]

4.

(a) What is a hash function, and why is it important in data structures? (5 points)

(b) Given an array of integers, describe how hash tables can be used to find if there are two numbers that sum up to a given value. (7 points)

(c) Compare Merge Sort and Quick Sort in terms of time complexity and practical use cases. (8 points)

TOTAL = [20 MARKS]